

MODEL QUESTION PAPER

TED (15) – 6214

Reg. No

REVISION – 2015

Signature

DIPLOMA EXAMINATION IN ENGINEERING / TECHNOLOGY /
MANAGEMENT / COMMERCIAL PRACTICE
AIRCRAFT INSTRUMENTS

[Time : 3 hours]

(Maximum marks : 100)

PART – A

(Maximum marks : 10)

Marks

I. Answer **all** questions in one or two sentences. Each question carries 2 marks.

1. List flight and navigation instruments.
2. Define Mach number.
3. Define degrees of freedom of Gyroscope.
4. Define dynamic pressure.
5. Define the working principle of thermocouple.

(5x2=10)

PART – B

(Maximum marks : 30)

II. Answer **any five** of the following questions. Each question carries 6 marks.

1. Explain the basic elements in an Aircraft instruments.
2. Describe the working of pitot static probe used in aircraft instruments.
3. Define pitch, bank and turn.
4. With figure explain immersion type thermocouple.
5. Explain the basic requirements of Aircraft instruments.
6. Describe the sensing and transmission method of pitot pressure.
7. Explain mach number and critical mach number.

(5x6=30)

PART – C

(Maximum marks : 60)

(Answer **one full question** from each unit. Each full question carries 15 marks.)

UNIT -- I

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| III. | (a) Describe gear mechanism in indicating instruments. | 8 |
| | (b) Describe lever mechanism with necessary diagrams. | 7 |

OR

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| IV. | (a) Explain hairspring and its controlling application in instruments. | 8 |
| | (b) Describe straight scale display and digital display. | 7 |

UNIT -- II

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|----|---|---|
| V. | (a) Describe the working of vertical speed indicator. | 7 |
| | (b) With neat diagram explain aneroid barometer. | 8 |

OR

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| VI. | (a) Write short notes on pitot pressure. | 8 |
| | (b) With neat sketch, describe the working of Air speed indicator. | 7 |

UNIT -- III

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|------|---|---|
| VII. | (a) Explain electric method of driving gyroscope rotor. | 8 |
| | (b) Explain tacho probe for engine speed measurement. | 7 |

OR

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|-------|--|---|
| VIII. | (a) Explain pneumatic method of driving gyroscope rotor. | 9 |
| | (b) Explain gyro horizon. | 6 |

UNIT -- IV

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|-----|--|---|
| IX. | (a) Describe the working of capacitance type fuel gauge used in aircrafts. | 8 |
| | (b) Explain trace recording. | 7 |

OR

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|----|---|---|
| X. | (a) With neat sketch explain the construction and working principle of accelerometer used in aircraft | 8 |
| | (b) With neat diagram explain the working of inductor pressure transmitter. | 7 |